

Notes: David, Hopenhayn, and Venkateswaran (QJE, 2016)

Information, Misallocation, and Aggregate Productivity

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1 Big picture

- **Question.** Can imperfect information at the firm level generate economically important misallocation, lower aggregate productivity, and explain part of the productivity gap between the United States, China, and India?
- **Core idea.** Firms choose inputs before perfectly observing their idiosyncratic fundamentals. Because they allocate capital (and possibly labor) using noisy signals, ex post marginal products are not equalized across firms.
- **Main theoretical contribution.** The paper embeds noisy firm learning and noisy stock-market learning into a standard heterogeneous-firm general equilibrium model. The key sufficient statistic is the posterior variance of the firm's fundamental, denoted by V : higher V means more residual uncertainty and more misallocation.
- **Main empirical contribution.** The paper proposes an identification strategy that combines production-side moments with stock-market moments. It uses the correlation of stock returns with future fundamentals and with investment, plus return volatility, to infer the information structure.
- **Headline result.** Informational frictions are quantitatively large. In the preferred conservative case where only capital is chosen under imperfect information, estimated TFP losses are about 4% in the United States, 7% in China, and 10% in India; corresponding output losses are about 5%, 10%, and 14%.
- **Mechanism result.** Learning from private information inside the firm matters much more than learning from stock prices. Even in the United States, stock-market information explains only a small share of total learning.
- **Why this paper matters.** The paper provides a concrete micro-foundation for misallocation. Instead of treating wedges as black boxes, it links them to a measurable economic friction: imperfect information.

2 Data and measurement (key objects)

2.1 Countries, firms, and periodization

- **Countries.** United States, China, and India.
- **Data sources.** Compustat North America for the United States and Compustat Global for China and India.
- **Sample.** Publicly traded firms; the quantitative exercise focuses on a 2012 cross section, using enough history to build nonoverlapping 3-year growth rates.
- **Time unit.** One period is interpreted as **three years**. This matches the paper's medium-run focus and helps avoid making investment look too sluggish simply because one uses very short horizons.
- **Why public firms?** The identification uses stock returns, so listed firms are essential. It also means the estimates likely understate informational frictions in the whole economy, since large public firms should be relatively well informed.

2.2 Fundamentals, investment, and prices

- **Firm fundamental.** The relevant idiosyncratic state is a demand/productivity shifter A_{it} , with log process

$$a_{it} = (1 - \rho)\bar{a} + \rho a_{i,t-1} + \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, \sigma_\varepsilon^2).$$

- **Observed proxy for fundamentals.** In the empirical implementation, the paper recovers the firm-level fundamental from production data as log value added minus a capital term implied by the model.
- **Investment moment.** ρ_{pi} is the cross-sectional correlation between lagged stock returns and subsequent investment growth.
- **Fundamental moment.** ρ_{pa} is the cross-sectional correlation between lagged stock returns and subsequent changes in fundamentals.
- **Price informativeness moment.** σ_p^2 is the variance of stock returns. Returns are unlevered and demeaned by time period so the identifying variation is cross-sectional.
- **Outlier treatment.** The paper trims the 2% tails of the empirical series.

2.3 Additional validation data

- **I/B/E/S Guidance.** Used to build a direct sector-level indicator of micro-uncertainty from firms' own forecast errors.
- **Jurado-Ludvigson-Ng micro-uncertainty.** Used to compare the paper's time-series measure of micro uncertainty in the United States to an external benchmark.

3 Models and empirical specifications (all equations + interpretation)

3.1 Production side

Intermediate-good firm i produces with capital and labor:

$$Y_{it} = K_{it}^{\hat{\alpha}_1} N_{it}^{\hat{\alpha}_2}, \quad (1)$$

with $\hat{\alpha}_1 + \hat{\alpha}_2 \leq 1$. Final output is a CES aggregate over intermediate goods with elasticity of substitution θ . This implies firm revenue

$$P_{it}Y_{it} = Y_t^{1/\theta} A_{it} K_{it}^{\alpha_1} N_{it}^{\alpha_2}, \quad (2)$$

where A_{it} is the firm-specific fundamental.

- **Interpretation.** The paper can read A_{it} either as a demand shifter or as a productivity shifter. For identification, this distinction is not crucial.
- **Key force.** The firm does not observe A_{it} perfectly when choosing inputs.

3.2 Case 1: both capital and labor chosen under imperfect information

The firm solves

$$\max_{K_{it}, N_{it}} Y_t^{1/\theta} \mathbb{E}_{it}[A_{it}] K_{it}^{\alpha_1} N_{it}^{\alpha_2} - W_t N_{it} - R_t K_{it}. \quad (3)$$

Using optimal factor proportions, the problem collapses to a capital-choice problem of the form

$$\max_{K_{it}} \text{constant} \times \mathbb{E}_{it}[A_{it}] K_{it}^\eta - R_t K_{it}, \quad (4)$$

where

$$\eta = \alpha_1 + \alpha_2 = (\hat{\alpha}_1 + \hat{\alpha}_2) \frac{\theta - 1}{\theta}. \quad (5)$$

Equilibrium capital demand is

$$K_{it} = \frac{(\mathbb{E}_{it}[A_{it}])^{1/(1-\eta)}}{\int (\mathbb{E}_{it}[A_{it}])^{1/(1-\eta)} di} K_t. \quad (6)$$

- **Interpretation.** Firms that expect stronger fundamentals choose more capital, but because expectations are noisy, ex post capital is misallocated.

3.3 Case 2: only capital chosen under imperfect information

Capital is chosen before the shock is perfectly learned, but labor is chosen after the current state is revealed. This yields a transformed capital problem:

$$\max_{K_{it}} \text{constant} \times \mathbb{E}_{it}[\tilde{A}_{it}] K_{it}^{\tilde{\eta}} - R_t K_{it}, \quad (7)$$

with

$$\tilde{A}_{it} = A_{it}^{1/(1-\alpha_2)}, \quad \tilde{\eta} = \frac{\alpha_1}{1 - \alpha_2}. \quad (8)$$

- **Interpretation.** This is the paper's conservative case. Firms can partially offset capital misallocation by adjusting labor after learning more.

3.4 Uncertainty as the sufficient statistic

If the firm's posterior over the fundamental is Gaussian,

$$a_{it} \mid I_{it} \sim N(\mathbb{E}_{it}[a_{it}], V), \quad (9)$$

then the posterior variance V is the key measure of residual uncertainty. In case 1,

$$\sigma_{\text{MRPK}}^2 = V. \quad (10)$$

- **Interpretation.** The forecast error variance maps directly into dispersion in marginal revenue products.
- **Core message.** Larger V means more informational misallocation.

3.5 Aggregate productivity and output losses

With constant returns to scale, aggregate output is

$$y = a + \hat{\alpha}_1 k + \hat{\alpha}_2 n. \quad (11)$$

Aggregate TFP is reduced by uncertainty:

$$\text{Case 1:} \quad a = \bar{a} - \frac{1}{2}\theta V, \quad (12)$$

$$\text{Case 2:} \quad a = \bar{a} - \frac{1}{2}(\theta\hat{\alpha}_1 + \hat{\alpha}_2)\hat{\alpha}_1\tilde{V}. \quad (13)$$

Allowing lower productivity to feed back into capital accumulation gives

$$\frac{dy}{dV} = \frac{da}{dV} \frac{1}{1 - \hat{\alpha}_1}, \quad (14)$$

with the analogous expression in case 2.

- **Interpretation.** Uncertainty lowers TFP directly through misallocation and lowers output further by depressing steady-state capital.
- **Case comparison.** Losses are larger in case 1 because labor cannot undo capital mistakes.

3.6 Signals and firm learning

The firm sees two noisy signals:

$$s_{it} = a_{it} + e_{it}, \quad e_{it} \sim N(0, \sigma_e^2), \quad (15)$$

and a stock-price signal that is informationally equivalent to

$$\hat{s}_{it} = a_{it} + \sigma_v z_{it}, \quad z_{it} \sim N(0, \sigma_z^2). \quad (16)$$

The posterior mean is a precision-weighted average,

$$\mathbb{E}_{it}[a_{it}] = V \left(\frac{(1 - \rho)\bar{a} + \rho a_{i,t-1}}{\sigma_\varepsilon^2} + \frac{s_{it}}{\sigma_e^2} + \frac{\hat{s}_{it}}{\sigma_v^2 \sigma_z^2} \right), \quad (17)$$

and the posterior variance is

$$V = \left(\frac{1}{\sigma_\varepsilon^2} + \frac{1}{\sigma_e^2} + \frac{1}{\sigma_v^2 \sigma_z^2} \right)^{-1}. \quad (18)$$

- **Interpretation.** More precise firm signals or more informative stock prices lower V .
- **Key decomposition.** The paper later uses this expression to separate learning from private information and learning from markets.

3.7 Stock market microfoundation

A noisy rational expectations stock market aggregates information from informed investors and noise traders. Noise traders buy a random amount $\Phi(z_{it})$, and the equilibrium threshold signal is

$$\hat{s}_{it} = a_{it} + \sigma_v z_{it}. \quad (19)$$

- **Interpretation.** Price informativeness depends on both investor signal precision and noise-trader noise.
- **Why it matters.** This gives the empirical meaning of $\sigma_v^2 \sigma_z^2$: it is the noise-to-signal ratio embedded in stock prices.

3.8 Other distortions and why naive misallocation measures are not enough

The paper allows capital-choice distortions that can be correlated or uncorrelated with fundamentals. Then observed dispersion in marginal products becomes

$$\sigma_{\text{MRPK}}^2 = V + \lambda^2(\sigma_a^2 - V) + \sigma_\varepsilon^2, \quad (20)$$

and the covariance between fundamentals and input choice becomes

$$\sigma_{ak} = \frac{1 + \lambda}{1 - \eta}(\sigma_\varepsilon^2 - V). \quad (21)$$

- **Interpretation.** Pure production-side moments confound uncertainty with other wedges.
- **Identification implication.** One cannot infer informational frictions directly from σ_{MRPK}^2 alone.

4 Identification and instruments (what makes the estimates credible)

4.1 Core identification idea

The paper uses stock prices as an observed component of the firm's information set. The logic is:

- if stock returns are very correlated with future fundamentals, prices are informative;
- if stock returns are very correlated with investment, firms react strongly to the information in prices;
- holding price informativeness fixed, a stronger return-investment correlation means firms are more uncertain and rely more on market information.

4.2 i.i.d. fundamentals case

When fundamentals are i.i.d., the paper derives closed-form expressions:

$$\rho_{pa} = \text{Corr}(\Delta p_{it}, \Delta a_{it}) = \left(1 + \frac{\sigma_v^2 \sigma_z^2}{\sigma_\varepsilon^2}\right)^{-1/2}, \quad (22)$$

$$\rho_{pk} = \text{Corr}(\Delta p_{it}, \Delta k_{it}) = \left(1 + \frac{\sigma_v^2 \sigma_z^2}{\sigma_\varepsilon^2}\right)^{-1/2} \left(1 - \frac{V}{\sigma_\varepsilon^2}\right)^{1/2}. \quad (23)$$

Combining them yields

$$\frac{V}{\sigma_\varepsilon^2} = 1 - \left(\frac{\rho_{pa}}{\rho_{pk}}\right)^2. \quad (24)$$

- **Interpretation.** The relative correlation ρ_{pa}/ρ_{pk} is the paper's sharpest sufficient statistic in the simple case.
- **Key intuition.** A high ρ_{pk} alone does not prove firms are uncertain. It can also arise because prices are very informative. ρ_{pa} isolates that channel.

4.3 Random walk case

When fundamentals follow a random walk, the same basic intuition survives, but the variance of returns is also needed. The paper derives a three-moment mapping between

$$(\rho_{pk}, \rho_{pa}, \sigma_p^2)$$

and the information parameters.

- **Interpretation.** Persistence complicates the algebra, but not the economic idea: more informative prices raise ρ_{pa} ; more uncertainty raises the relative comovement of investment with prices.

4.4 Why the strategy is robust

- **Correlated distortions.** If non-informational distortions simply scale the firm's responsiveness to fundamentals, correlations are unchanged, so the inferred V is unchanged.
- **Uncorrelated distortions.** These bias the estimate downward, making the method conservative.
- **Correlated firm and market information.** Even if stock prices contain information already known to the firm, the special cases still recover the true V .
- **Measurement error.** Revenue mismeasurement tends to bias V upward; capital mismeasurement has ambiguous sign analytically, but in the quantitative exercises often biases V downward.

5 Tables and figures

5.1 Figure 1 and Table 1: quantitative identification setup

Read:

- Table 1 sets the calibration backbone: 3-year periods, discount factor 0.9, capital share 0.33, labor share 0.67, and baseline elasticity of substitution $\theta = 6$.

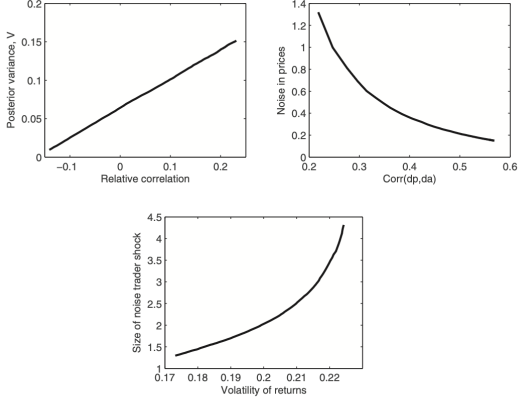


FIGURE 1
Identification—Quantitative Model

TABLE I PARAMETERIZATION: SUMMARY		
Parameter	Description	Target/Value
Common		
	Time period	3 years
β	Discount rate	0.90
$\hat{\alpha}_1$	Capital share	0.33
$\hat{\alpha}_2$	Labor share	0.67
θ	Elasticity of substitution	6
Country-specific		
ρ	Persistence of fundamentals	Estimates of eq. (1): $a_{it} = (1 - \rho)\bar{a} + \rho a_{it-1} + \mu_{it}$
σ_a	Shocks to fundamentals	
σ_e^2	Firm private signal	
σ_v^2	Investor private signal	
σ_z^2	Noise trading	
		ρ_{pi}
		ρ_{pa}
		σ_p^2

- It also clarifies the empirical mapping: ρ and σ_e come from the AR(1) in fundamentals, while σ_e , σ_v , and σ_z are disciplined by ρ_{pi} , ρ_{pa} , and σ_p^2 .
- Figure 1 shows the identification geometry in the quantitative model: V rises with the relative correlation, noise in prices falls as ρ_{pa} rises, and higher return volatility implies larger noise-trader shocks.

5.2 Table 2: target moments and estimated parameters

Read:

- The target moments vary strongly across countries. In case 2, ρ_{pi} is 0.23 in the United States, 0.16 in China, and 0.25 in India; ρ_{pa} is highest in the United States at 0.18 and much lower in China and India.
- The parameter estimates imply cleaner information in the United States: lower private-signal noise, lower investor-signal noise, and lower noise-trader noise.
- The paper's interpretation is that U.S. firms face both better private information and better market information, while China and India also face more volatile fundamentals.

TABLE II TARGET MOMENTS AND PARAMETER ESTIMATES								
	Target Moments			Parameters				
	ρ_{pi}	ρ_{pa}	σ_p^2	ρ	σ_μ	σ_e	σ_v	σ_z
Case 2								
United States	0.23	0.18	0.23	0.92	0.45	0.39	0.37	3.50
China	0.16	0.06	0.14	0.78	0.51	0.67	0.74	4.24
India	0.25	0.08	0.23	0.93	0.53	1.04	0.69	4.36
Case 1								
United States	0.24	0.10	0.23	0.88	0.46	0.63	0.65	3.16
China	0.15	0.02	0.14	0.75	0.53	0.74	1.18	3.14
India	0.26	0.00	0.22	0.88	0.55	1.39	1.69	4.14

TABLE III THE IMPACT OF INFORMATIONAL FRICTIONS						
	V	$\frac{V}{\sigma_e^2}$	σ_{mrpk}^2	$\frac{V}{\sigma_{mrpk}^2}$	$\alpha^* - \alpha$	$y^* - y$
Case 2 ($\alpha = 0.62$)						
United States	0.08	0.41	0.38	0.22	0.04	0.05
China	0.16	0.63	0.47	0.34	0.07	0.10
India	0.22	0.77	0.46	0.48	0.10	0.14
Case 1 ($\alpha = 0.83$)						
United States	0.13	0.63	0.38	0.35	0.40	0.60
China	0.18	0.65	0.47	0.39	0.55	0.83
India	0.26	0.86	0.46	0.56	0.77	1.15

5.3 Table 3: baseline quantitative impact of informational frictions

Read:

- Residual uncertainty is large. In case 2, V/σ_ε^2 equals 0.41 in the United States, 0.63 in China, and 0.77 in India.
- Informational frictions account for a sizable share of observed MRPK dispersion: 22%, 34%, and 48% in case 2 for the United States, China, and India, respectively.
- Aggregate consequences are economically important. In case 2, TFP losses are 4%, 7%, and 10%; output losses are 5%, 10%, and 14%. In case 1, losses are much larger: 40%, 55%, and 77% for TFP, and 60%, 83%, and 115% for output.
- The paper treats case 2 as a lower bound and case 1 as an upper bound.

5.4 Tables 4 and 5: curvature and total learning

TABLE IV
THE IMPACT OF INFORMATIONAL FRICTIONS: ALTERNATIVE VALUES OF θ

	$\theta = 4$					$\theta = 10$				
	V	$\frac{V}{\sigma_\mu^2}$	$\frac{V}{\sigma_{mrpk}^2}$	$\alpha^* - \alpha$	$y^* - y$	V	$\frac{V}{\sigma_\mu^2}$	$\frac{V}{\sigma_{mrpk}^2}$	$\alpha^* - \alpha$	$y^* - y$
United States	0.05	0.25	0.14	0.02	0.03	0.12	0.57	0.31	0.08	0.11
China	0.13	0.52	0.28	0.04	0.06	0.17	0.63	0.36	0.11	0.17
India	0.17	0.60	0.37	0.06	0.08	0.24	0.83	0.53	0.16	0.24

TABLE V
THE IMPORTANCE AND SOURCES OF LEARNING

	ΔV	$\frac{\Delta V}{\sigma_\mu^2}$	$\Delta \alpha$	Δy	Share from Source	
					Private	Market
Case 2						
United States	-0.12	-0.59	0.05	0.08	0.92	0.08
China	-0.10	-0.37	0.04	0.06	0.96	0.04
India	-0.06	-0.23	0.03	0.04	0.89	0.11
Case 1						
United States	-0.08	-0.37	0.23	0.35	0.91	0.09
China	-0.10	-0.35	0.30	0.45	0.96	0.04
India	-0.04	-0.14	0.12	0.19	0.96	0.04

Notes. The left side shows the total effect of learning, that is, from all sources: the reduction in V and the corresponding gains in the aggregates (hence the pattern of negatives and positives). The right side shows the relative shares of private and market sources in total learning.

Read:

- Table 4 varies θ . A lower elasticity of substitution ($\theta = 4$) makes the aggregate cost of misallocation smaller; a higher one ($\theta = 10$) makes it much larger.
- This is economically intuitive: when goods are closer substitutes, inefficient allocation across firms hurts more.
- Table 5 decomposes how much total uncertainty is eliminated by learning from all sources combined. The total gains are substantial, but the decomposition already hints that private information dominates market information.

5.5 Table 6: market information versus private information

Read:

- Market information contributes little. Even when it is the only source of learning, it reduces uncertainty by only 5% to 10% of underlying uncertainty in the United States and even less in China and India.
- Private information contributes much more. In the presence of markets, it removes about 32% to 57% of uncertainty in the United States and still large shares in China and India.
- The paper's RPE versus FPE interpretation is important: stock prices have low revelatory price efficiency mainly because they have low forecasting price efficiency.

TABLE VI
THE CONTRIBUTIONS OF MARKET AND PRIVATE INFORMATION

	Market Information				Private Information			
	Both Sources		Only Market		Both Sources		Only Private	
	$\frac{\Delta V}{\sigma_\mu^2}$	Δa	$\frac{\Delta V}{\sigma_\mu^2}$	Δa	$\frac{\Delta V}{\sigma_\mu^2}$	Δa	$\frac{\Delta V}{\sigma_\mu^2}$	Δa
Case 2								
United States	-0.021	0.002	-0.107	0.009	-0.48	0.04	-0.57	0.05
China	-0.010	0.001	-0.025	0.003	-0.35	0.04	-0.36	0.04
India	-0.019	0.002	-0.030	0.004	-0.20	0.02	-0.21	0.03
Case 1								
United States	-0.021	0.013	-0.051	0.031	-0.32	0.20	-0.37	0.22
China	-0.009	0.008	-0.022	0.017	-0.33	0.28	-0.38	0.29
India	-0.005	0.004	-0.006	0.005	-0.13	0.12	-0.14	0.12

Notes. The table shows the reductions in V due to market and private information and the corresponding gains in aggregate TFP (hence the pattern of negatives and positives). Columns under Both Sources show the effects of each source (market/private) of learning in the presence of the other. Columns under Only Market and Only Private show the effect of each when it is the only source of information.

TABLE VII
THE EFFECTS OF A U.S. INFORMATION STRUCTURE

	Case 2				Case 1			
	ΔV	$\frac{\Delta V}{\sigma_\mu^2}$	Δa	Δy	ΔV	$\frac{\Delta V}{\sigma_\mu^2}$	Δa	Δy
Market information								
China	-0.01	-0.05	0.01	0.01	-0.01	-0.02	0.02	0.02
India	-0.02	-0.07	0.01	0.01	-0.01	-0.05	0.04	0.06
Private information								
China	-0.07	-0.26	0.03	0.04	-0.02	-0.08	0.06	0.09
India	-0.12	-0.43	0.05	0.08	-0.09	-0.31	0.26	0.39
Shocks								
China	-0.02	-0.09	0.01	0.02	-0.03	-0.13	0.10	0.15
India	-0.05	-0.19	0.02	0.03	-0.07	-0.24	0.20	0.31

Notes. The table shows the reduction in V and the corresponding gains in the aggregates (hence the pattern of negatives and positives) in India and China from having access to U.S.-quality financial market information (top panel), private information (middle panel) and facing U.S. fundamental shocks (bottom panel).

5.6 Table 7: cross-country counterfactuals

Read:

- Giving China and India U.S.-quality stock-market information produces only modest gains.
- Giving them U.S.-quality *private* information produces much larger gains: in some cases output rises by about 8% in case 2 and nearly 40% in case 1.
- Replacing Chinese and Indian fundamental shocks with U.S.-style less volatile shocks also matters, implying that higher volatility itself makes inference harder in emerging markets.

5.7 Tables 8, 9, and 10: robustness

TABLE VIII
UNCERTAINTY WITH CORRELATED INFORMATION

	σ_ϵ	σ_v	σ_π	V	Baseline V
United States	0.37	0.17	8.01	0.08	0.08
China	0.60	0.63	4.65	0.15	0.16
India	0.79	0.45	6.37	0.19	0.22

Notes. The table shows parameter estimates and the resulting extent of uncertainty when investors are assumed to observe a noisy signal of the firm's private signal. All parameters not displayed retain the same values as the baseline case.

TABLE IX
A FULL-INFORMATION ADJUSTMENT COST MODEL

	Relative Correlation		Implied Uncertainty		Baseline Uncertainty	
	Data	Model	V	$a^* - a$	V	$a^* - a$
United States	0.05	-0.03	0.03	0.01	0.08	0.04
China	0.10	-0.08	0.06	0.03	0.16	0.07
India	0.17	-0.02	0.08	0.04	0.22	0.10

Notes. The table shows the relative correlation of returns with investment growth and returns with fundamentals ($\rho_{\mu_t} - \rho_{\pi_t}$) as computed from the data and as predicted by the full-information adjustment cost model, as well as the implied extent of uncertainty from the latter and from the baseline model. ρ_{μ_t} is a targeted moment and so is the same in the data and model.

Read:

- Table 8 allows firm and market signals to be correlated. Estimated uncertainty barely changes relative to baseline, confirming the identification strategy is not driven by the conditional-independence assumption.
- Table 9 asks whether a full-information model with capital adjustment costs can mimic the data. It cannot. The implied uncertainty and TFP losses are only about one-third of baseline, so adjustment costs alone do not explain the moments.
- Table 10 studies measurement error. Revenue mismeasurement tends to inflate the estimated V , especially in the United States; capital mismeasurement tends to push V upward or downward depending on the case, but the cross-country ranking remains broadly robust.

TABLE X
THE EFFECT OF MEASUREMENT ERROR

	Baseline		Error in y 10%		Error in y 25%		Error in k 10%	
	V	$\frac{V}{\sigma_\mu^2}$	V	$\frac{V}{\sigma_\mu^2}$	V	$\frac{V}{\sigma_\mu^2}$	V	$\frac{V}{\sigma_\mu^2}$
Case 2								
United States	0.08	0.41	0.05	0.28	0.01	0.07	0.11	0.59
China	0.16	0.63	0.14	0.58	0.10	0.51	0.17	0.69
India	0.22	0.77	0.17	0.65	0.14	0.61	0.22	0.79
Case 1								
United States	0.13	0.63	0.12	0.62	0.07	0.42	0.17	0.85
China	0.18	0.65	0.18	0.69	0.17	0.73	0.21	0.79
India	0.26	0.86	0.25	0.91	0.22	0.89	0.27	0.95

5.8 Figures 2 and 3: validating the uncertainty measure

Read:

- Figure 2 compares sectoral estimates of V to sectoral forecast errors from I/B/E/S Guidance. Sectors with larger direct forecast errors also have higher model-implied uncertainty.
- Figure 3 builds a U.S. time series of micro uncertainty. The series is countercyclical and rises in recessions, broadly lining up with the Jurado-Ludvigson-Ng measure.
- These figures matter because uncertainty is not directly observed, so external validation is valuable.

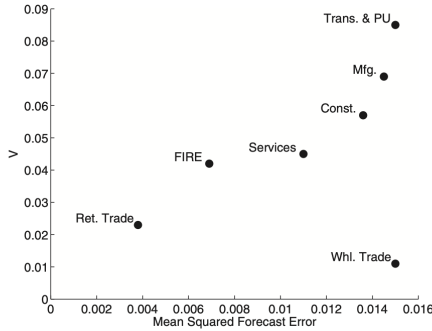


FIGURE II

Uncertainty at the Sectoral Level

The top panel displays annual estimates of V . The bottom panel shows annual estimates of V and annual estimates of micro-uncertainty from Jurado, Ludvigson, and Ng (2013). All series are detrended and normalized by their standard deviation.

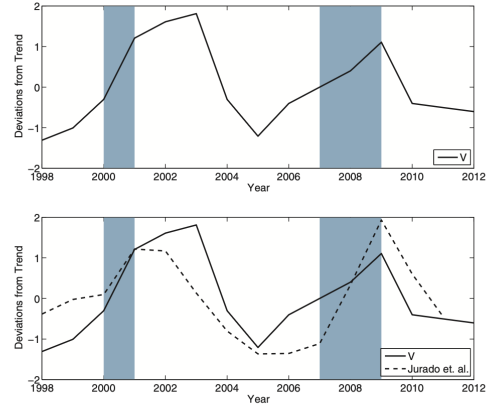


FIGURE III

Uncertainty through Time

6 Short summary

- **Conclusion.** Imperfect information generates meaningful misallocation and nontrivial losses in productivity and output.
- **Mechanism.** Firms choose inputs using noisy signals about their fundamentals; stock prices help a little, but internal firm information helps much more.
- **Empirical lesson.** The relative comovement of returns with fundamentals and investment can isolate informational frictions even when other wedges contaminate standard misallocation moments.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** A large fraction of observed cross-firm marginal-product dispersion is consistent with imperfect information, not only with taxes, financial frictions, or adjustment costs.
- **Magnitude.** In the conservative capital-only case, the estimated losses are about 4% of TFP in the United States and up to 10% in India.
- **Credibility.** The identification combines production moments with stock-market moments and is robust to many classes of omitted distortions.
- **Mechanism evidence.** Internal firm information is far more important than stock prices, and the uncertainty series behaves like a plausible measure of micro uncertainty.
- **Cross-country implication.** Differences in firm private information quality and in the volatility of fundamentals appear more important than stock-market information in explaining why informational frictions are worse in China and India.

7.2 Economic intuition

- Firms do not equalize marginal products when they do not know their own current demand or productivity well enough at the time of input choice.
- Stock prices aggregate information, but only imperfectly; if prices are noisy, firms cannot lean heavily on them.
- Better internal information or lower volatility reduce forecast errors, improve the allocation of capital, and raise aggregate productivity.

7.3 Takeaway.

The paper turns “misallocation” from a residual into a measurable informational friction. Its main lesson is that *firm-level informational quality* is quantitatively important for resource allocation and aggregate performance, so improving managerial information and internal data processing may matter at least as much as improving financial-market informativeness.